

Slide 1

HOW DOES THE INTERNET ACTUALLY WORK?

OSI & TCP/IP Explained

What really happens after you press SEND?

Slide 2

HOW DOES THE INTERNET ACTUALLY WORK?

You Send “Hey 🙋”... Then What?

Chat Prompt: What happens after you press SEND on WhatsApp?

Slide 3

Today's Mission

- Understand networking layers
- Learn the OSI model
- Understand TCP/IP
- Compare TCP vs UDP
- Trace real internet communication

Slide 4

Why Do Networks Use Layers?

Restaurant Analogy 🍔

- ▸ Waiter takes the order
- ▸ Chef cooks the food
- ▸ Cashier handles payment
- ▸ Delivery rider transports food
- ▸ Each role has ONE responsibility

Slide 5

The Journey of Data

- ▸ Application creates data
- ▸ Transport manages delivery
- ▸ Network handles routing
- ▸ Physical layer transmits signals
- ▸ Destination reverses the process

Slide 6

The OSI Model

- ▸ 7. Application
- ▸ 6. Presentation
- ▸ 5. Session
- ▸ 4. Transport
- ▸ 3. Network
- ▸ 2. Data Link
- ▸ 1. Physical

Slide 7

Memory Trick 🤔

Please Do Not Throw Sausage Pizza Away

Slide 8

Application Layer

Where humans touch the network

- ▸ Browsers
- ▸ WhatsApp
- ▸ YouTube
- ▸ HTTP / HTTPS / DNS

Slide 9

Presentation Layer

The Translator

- ▸ Encryption
- ▸ Compression
- ▸ Formatting
- ▸ Makes devices understand each other

Slide 10

Session Layer

Keeps conversations alive

- ▶ Zoom meetings
- ▶ Login sessions
- ▶ Gaming sessions
- ▶ Opens and closes communication

Slide 11

Transport Layer

TCP vs UDP 🔥

TCP

- ✓ Reliable
- ✓ Ordered delivery
- ✓ Slower
- ✓ Email & Banking

UDP

- ⚡ Faster
- ⚡ No guarantees
- ⚡ Low latency
- ⚡ Gaming & Streaming

Slide 12

Quick Poll

- Netflix → TCP or UDP?
- Gaming → TCP or UDP?
- Online Banking → TCP or UDP?
- Voice Calls → TCP or UDP?

Slide 13

Network Layer

The GPS of the Internet 🌐

- ▸ IP addresses
- ▸ Routing packets
- ▸ Routers
- ▸ Decides WHERE data goes

Slide 14

Data Link Layer

Local Delivery

- ▸ MAC addresses
- ▸ Switches
- ▸ Local communication
- ▸ Apartment analogy: IP vs MAC

Slide 15

Physical Layer

Signals & Cables ⚡

- ▸ Ethernet cables
- ▸ Fiber optics
- ▸ WiFi signals
- ▸ Electrical / Light / Radio transmission

Slide 16

Guess The Layer

- ▷ HTTPS
- ▷ Router
- ▷ Fiber Cable
- ▷ MAC Address
- ▷ TCP
- ▷ IP Address

Slide 17

TCP/IP Model

The Real Internet Model 

- ▸ Application
- ▸ Transport
- ▸ Internet
- ▸ Network Access

LEARNING OBJECTIVES

Describe the four layer TCP/IP model.

Describe the functions of each layer in a network.

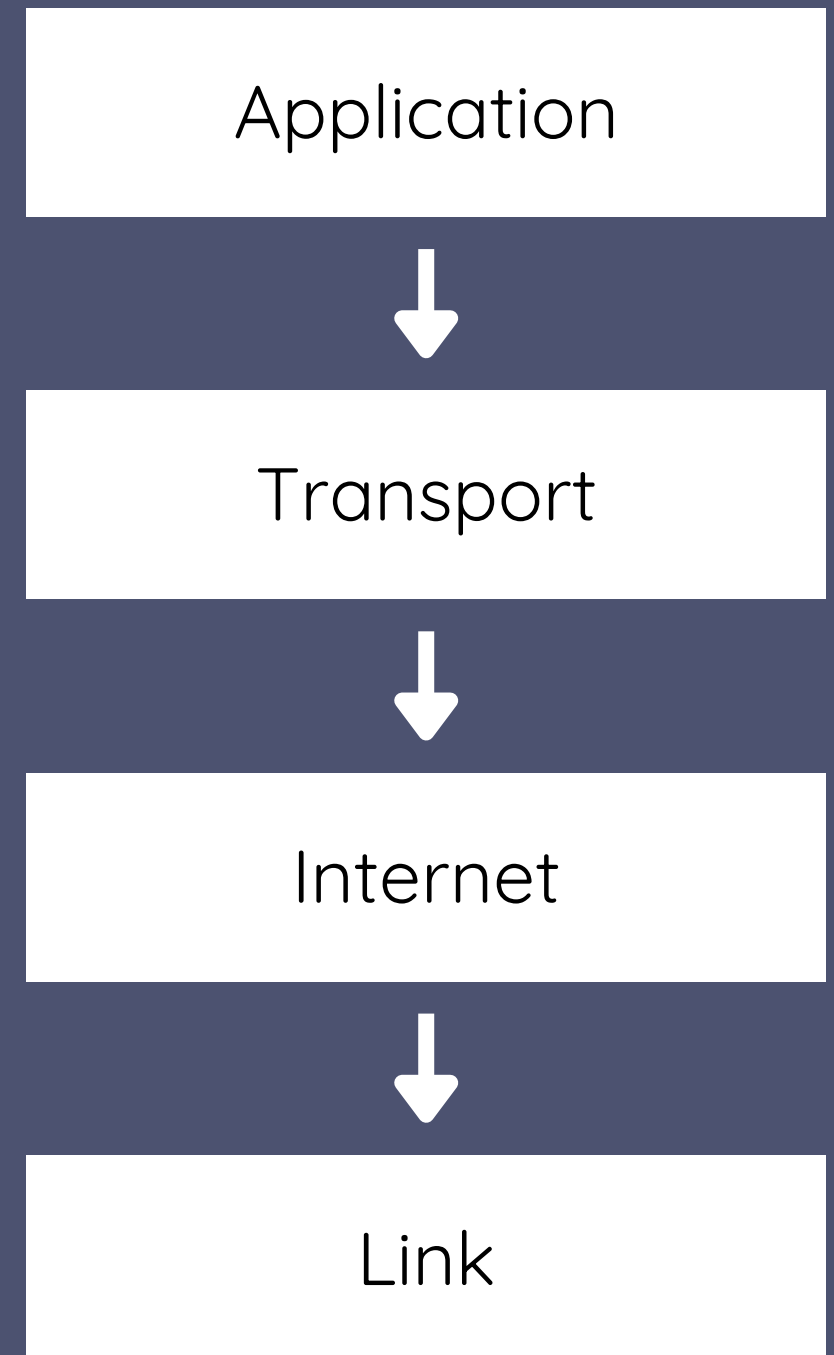
Understand the protocols that operate on each layer.

TCP/IP MODEL

Protocol means a rule that defines how devices will communicate with each other.

The internet uses a series of different protocols. These protocols are organised into a stack. This is called the TCP/IP model and has four layers that sit atop each other.

The layer of protocols exists for each device that is communicating.



APPLICATION

TRANSPORT

INTERNET

LINK

The highest layer of the TCP/IP model is where the application sits. The application sends or receives data that is then sent into the rest of the protocol stack.

The application layer adds header data that can be tracked through the stack.

Various different protocols sit on the application layer, depending on the type of application being used.

Examples of protocols:

- HTTP
- HTTPS
- SMTP
- IMAP
- FTP



APPLICATION

TRANSPORT

INTERNET

LINK

Data is passed from the application layer down to the transport layer to be broken into packets. Each packet of data is given a sequence number so that it can be reassembled.

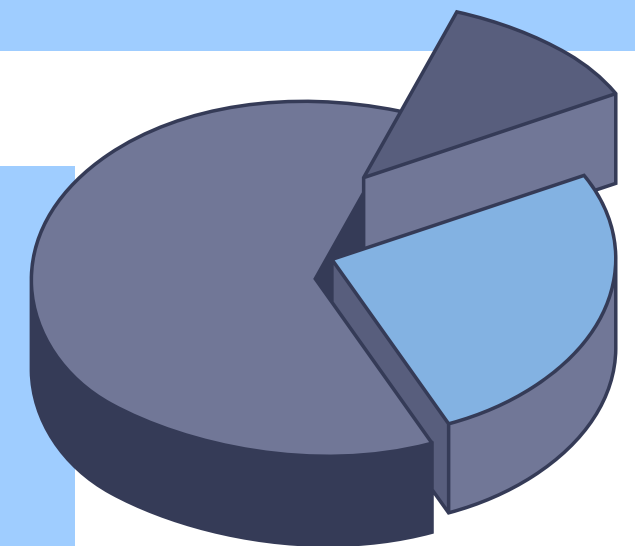
Source and destination ports are added to the header.

Transmission Control Protocol (TCP)

TCP is a reliable transport protocol because it will always request again any missing data packets.

User Datagram Protocol (UDP)

UDP is a faster transport protocol but this makes it unreliable because missing data packets are discarded.



Application

Transport

Internet

Link

The internet layer routes packets of data from one device to another across a network.

Each device is assigned a unique IP (Internet Protocol) address, which identifies the source and destination for data packets.

The internet layer is where the best route is determined for the data packet to get from its source IP address to its destination.



APPLICATION

TRANSPORT

INTERNET

LINK

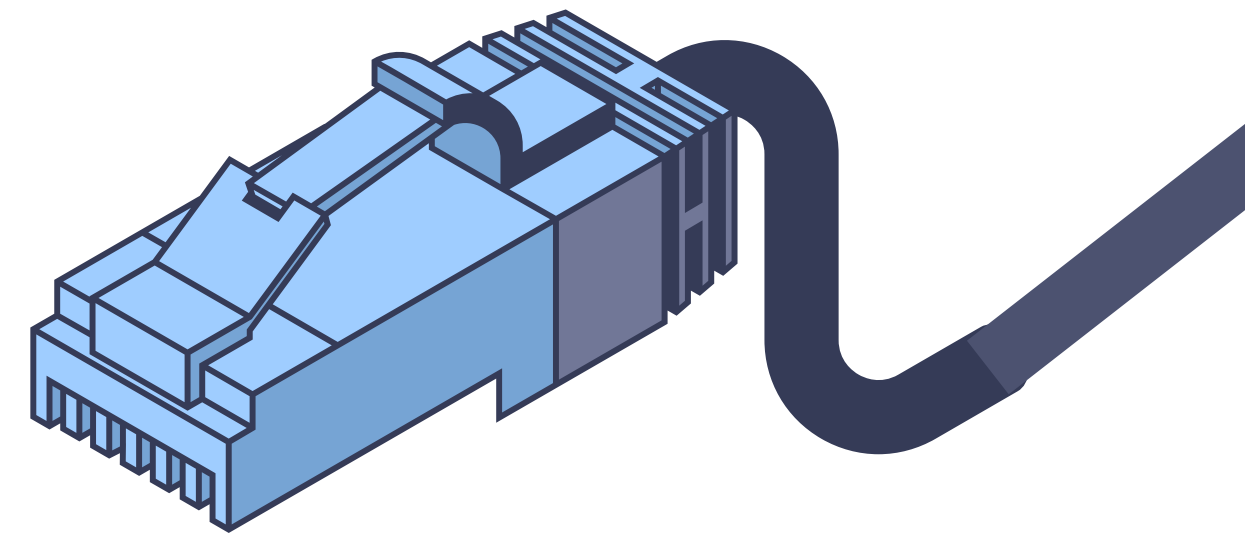
The link layer encompasses the protocols of different communicating technologies.

The most common of these protocols is Ethernet, which is used in local area networks. The protocol determines how packets of data are transmitted between devices.

There are a variety of different link layer protocols, some wired and others wireless.

Link layer protocols:

- Ethernet
- WiFi
- Fibre
- 4G
- Satellite



For each question, remove the incorrect answer.

Which protocol operates on the transport layer?

TCP

FTP

Which protocol operates on the application layer?

IP

HTTP

Which protocol operates on the link layer?

IMAP

Ethernet

Which protocol operates on the application layer?

FTP

WiFi

Which protocol operates on the transport layer?

UDP

SMTP

For each question, remove the incorrect answer.

Which protocol operates on the transport layer?

TCP

Which protocol operates on the application layer?

HTTP

Which protocol operates on the link layer?

Ethernet

Which protocol operates on the application layer?

FTP

Which protocol operates on the transport layer?

UDP

Packets Successfully Delivered...Thank you!